Exercise – *Alice and the Hurried Rabbit Clan*

"Oh dear! Oh dear! I shall be too late!"

The White Rabbit, Alice's Adventures in Wonderland *by Lewis Carroll (1832–1898)*

After being lost in the Tulgey Wood for what seems to be an eternity, Alice is stranded at the peak of a mountain and has but one option left. She must find the caterpillar and get his wise advice, otherwise there is nothing but demise. The caterpillar resides at the lowest point of the valley. But that's not all. The White Rabbit's family, the hurried rabbit clan, wants to see Alice! It is important to the hurried rabbit clan to be efficient. This means that they want to walk as little as possible and they walk east or south only, as it goes downhill that way. Even more so, it is important to them that also Alice walks east or south only. Nice as Alice is, she does not want to anger them and wants to help them walk as little as possible. Hence, she wants to find a path to the caterpillar that minimizes the distance the rabbits of the hurried rabbit clan have to walk. Luckily, the rabbits shared their location with Alice on their phone and she can let them know the path she plans to take before she starts walking.

The peak of the mountain, where Alice is located at the beginning, is at position $(0,0)$ of an $n \times m$ integer grid. The caterpillar awaits her at position $(n-1, m-1)$. Walking east or south means that any character in our story can go in one step from (i,j) to $(i+1,j)$ or to $(i,j+1)$, as long as they stay within the grid. For each position (i,j) on the grid, you know the number $r_{i,j}$ of rabbits located at (i,j). Given a path P that Alice takes to the caterpillar, each rabbit at (i,j) travels a distance of $d(i,j,P)$, which is the minimum number of east or south steps needed to get from (i,j) to some position on P. Alice selects her path P such that the total distance $\sum_{(i,j)\in n\times m} r_{i,j}\, d(i,j,P)$ traveled by all rabbits is minimal. So, what is this total distance?

Input The first line of the input contains the number $t \leqslant 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line that contains two integers n m, separated by a space. They denote
  - n, the number of rows in the grid ($1 \leqslant n \leqslant 800$) and
  - m, the number of columns in the grid ($1 \leqslant m \leqslant 800$).
- The following n lines each contain m integers, separated by a space, describing the number of rabbits at each position of the grid. That is, the j-th number on the i-th line denotes the number $r_{i,j}$ (where $0 \leqslant r_{i,j} \leqslant 10^8$) of rabbits at position (i,j).

Output For each test case output a single line with a single integer that denotes $\min_P \sum_{(i,j)\in n\times m} r_{i,j}\, d(i,j,P)$.

Points There are four groups of test sets, each of which is worth 20 points. For each group there is also a corresponding hidden test set that is worth 5 points. So, you can achieve $4 \cdot 20 + 4 \cdot 5 = 100$ points overall.

1. For the first group of test sets, you may assume that $m = 2$ and that there are exactly two different positions with a nonzero number of rabbits.
2. For the second group of test sets, you may assume that $m = 2$.

3. For the third group of test sets, you may assume that $n, m \leqslant 10^2$.

4. For the fourth group of test sets, there are no additional assumptions.

Sample Input

```
3
3 3
1 2 3
1 2 3
1 2 3
3 3
7 8 4
3 1 3
9 7 5
3 3
7 10 10
3 1 10
16 7 5
```

Sample Output

```
3
3 3
1 2 3
1 2 3
1 2 3
3 3
7 8 4
3 1 3
9 7 5
3 3
7 10 10
3 1 10
16 7 5
```